

WASP-2b

R-0571

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-2_230913 · created 2026-07-24T15:57:50+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460200.6693 +/- 0.0076 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.157 +/- 0.047
Transit depth (Rp/Rs)^2	2.47 +/- 1.48 [%]
Orbital Inclination (inc)	85.48 +/- 2.88
Airmass coefficient 1 (a1)	27.84 +/- 16.13
Airmass coefficient 2 (a2)	-2.92 +/- 1.4
Scatter in the residuals of the lightcurve fit is	57.38 %
Best Comparison Star	#3 - [562, 26]
Optimal Aperture	11.79
Optimal Annulus	12.77
Transit Duration (day)	0.058 +/- 0.025

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.157 $\pm$ 0.047 vs 0.1326 (deviation 0.42 σ)
Duration fit vs published	0.058 d vs 0.0737 d (deviation 0.52 σ)
Mid-time O-C	-9.3 min
Depth SNR	2.7
Residual scatter	57.38 %

Light curve

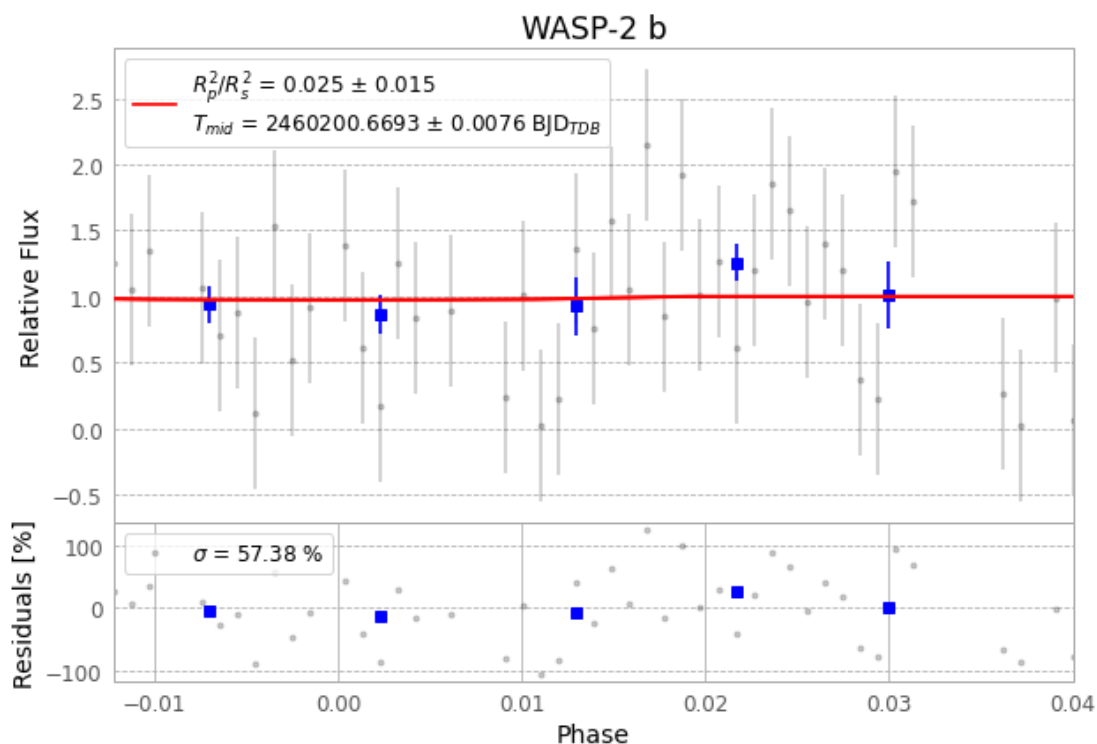


Figure 1 — FinalLightCurve_WASP-2 b_2023-09-12.png (SHA-256 eae593f27ff18c...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [465, 190], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 c182c69be6c24574...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit b365494

Data provenance

69 FITS frames (45 MB), WASP-2230913023315.FITS → WASP-2230913060315.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-2-230913.log (aeddb660f7a3...), FinalLightCurve_WASP-2 b_2023-09-12.png (eae593f27f...), FinalLightCurve_WASP-2 b_2023-09-12.csv (90bf6388023b...)

Compute resources

Wall time 210.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-24 15:57Z.